PARKER MACE

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EDUCATION

Tufts University, Medford, MA | B.S. Mechanical Engineering, Minor in Music Engineering | Expected 2028
GPA: 3.4 | Dean's List (all semesters) | Engineer of the Year, Newton North High School (2024)

ENGINEERING PROJECTS

Service Dog Treat Dispenser, Assistive Design | May 2026

- Designed and modeled the treat magazine for a dispenser built for a wheelchair user and his service dog.
- Designed a clear acrylic channel with a rubber-band gate that holds treats and lets one out when pushed, so he can get a treat easily from his chair; produced the assembly drawing and bill of materials.

Cam-Driven Embosser, Mechanism Design | March 2026

- Designed the mechanism for a motorized embosser with a partner: the drivetrain, the cam press, and the die loading system.
- Designed a 3:1 gear reduction with laser-cut gears and a cam that lifts a guided die holder into the die negative.
- Designed a layered top case with magnets holding the dies, so dies swap without taking the machine apart.

Crusader Helmet, Sheet Metal Fabrication | November 2025

- Led CAD for a five-person team: redrew a paper pattern by hand in SolidWorks and scaled it from real head measurements.
- Added rivet tabs after a waterjet test piece; helped cut, bend, drill, and rivet the aluminum panels.

Truss Analysis and Load Test, Structural Analysis | October 2025

- Calculated reactions and member forces by hand for a 7-joint, 11-member Warren truss under center and off-center loads, and checked them with analysis software.
- Built and load-tested a physical truss with a partner to compare against the predictions.

Wind Tunnel Design and Build, Team Lead | September 2023 – May 2024

- Led a team to design and build a 17-ft wind tunnel with a 1 ft × 1 ft test section, designed in Onshape for easy disassembly.
- Achieved laminar flow with a paper-tube flow straightener and confirmed it with smoke flow-visualization tests.
- Planned and coordinated fabrication across the team and finished on schedule.

SKILLS

CAD: SolidWorks, Onshape, Fusion 360 | **Simulation:** COMSOL Multiphysics (CFD) | **Programming:** MATLAB, Python, C#
Manufacturing: Manual mill, lathe, CNC, waterjet, laser cutter

EXPERIENCE

Landry's Bicycles, Sales and Repair Technician | Summer 2024: Repaired customer bikes and trained new employees.
Community Rowing Inc., Equipment and Grounds Volunteer | 2019 – 2024: Repaired rowing shells and equipment.
University Club, Lifeguard | Summer 2025: Kept the pool safe and taught swim lessons.

LEADERSHIP AND ACTIVITIES

Rocketry Team (TARC), Co-Founder and Co-President: founded and led a 10-member team that designed, built, and flew competition rockets; ran workshops and fundraising.
Tufts University Men's Crew, Varsity Rower | 2024 – Present. **Community Rowing Inc.**, Rower and Captain | 2018 – 2024